

Mapbox Navigation – On-Route and Off-Route Detection

1. Current Situation

Our application is currently using:

```
flutter_mapbox_navigation: ^0.2.2
```

We need to identify whether the user is:

- Following the planned route (On-Route)
- Moving away from the planned route (Off-Route)

2. Current Issue

The current Flutter package does not provide a direct feature to check whether the user is on or off the planned route.

For example, we cannot directly use something like:

```
isOffRoute = true
```

or:

```
onRoute / offRoute
```

Therefore, we need to add additional native Android code to access the required route detection functionality.

3. Why Is Native Android Code Required?

The Mapbox navigation system works internally with native Android code.

The current Flutter package acts as a connection between Flutter and the Mapbox navigation system. However, the package does not provide all the required features directly in Flutter.

Because of this, some features need to be accessed using native Android code and then connected back to the Flutter application.

In simple terms:

```
Flutter App
↓
Flutter Mapbox Package
↓
Native Android Mapbox Navigation
```

The required route detection functionality is available at the navigation system level, but it is not directly available through the current Flutter package.

4. Option 1: Add Native Android Code

We can add native Android code to get the required route status and connect it with Flutter.

Benefits

- More accurate route detection.
- Uses the Mapbox navigation system's own functionality.

- Better control over navigation features.
- Easier to support advanced navigation requirements in the future.

Limitation

- Additional Android native code will be required.
- Flutter and native Android code will need to communicate with each other.

5. Is There a Newer Version of the Current Package?

The current package is:

```
flutter_mapbox_navigation: ^0.2.2
```

Based on the current package status, flutter_mapbox_navigation: ^0.2.2 is still the latest stable version of this exact package. It was published approximately two years ago, and there is currently no newer stable version available in the 0.2.x series.

There is also no newer stable version of this same package that directly provides an onRoute or offRoute feature in Flutter.

Therefore, simply updating the current package does not solve the requirement for direct on-route and off-route detection.

Other Flutter packages may provide this feature, but each package needs to be evaluated carefully before migration.

6. Conclusion

The current Flutter package is suitable for basic navigation, but it does not directly provide the required on-route and off-route detection feature.

To achieve reliable route deviation detection, we need to add native Android code to access the required functionality.

For a long-term and more reliable solution, using the native Android Mapbox Navigation SDK is recommended, especially because the application requires advanced navigation and fleet management features.